STAT 565 Homework 5

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In [25]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
from statsmodels.tsa.arima.model import ARIMA

df = pd.read_csv('m-unrate.txt', sep=r'\s+', header=0)
df['date'] = pd.to_datetime(df[['Year', 'Mon', 'Day']].rename(columns={'Year': 'ye
xt = df.set_index('date')['Rate']

# y_t = Delta x_t
yt = xt.diff().dropna()
```

Part (a): AR(12) model

In [26]:
```
model_full = ARIMA(yt, order=(12, 0, 0))
fit_full = model_full.fit()

print("AR(12) Model Summary:")
print(fit_full.summary())
```

```
AR(12) Model Summary:
                               SARIMAX Results
==============================================================================
Dep. Variable:                   Rate   No. Observations:                  734
Model:                ARIMA(12, 0, 0)   Log Likelihood                 153.294
Date:                Mon, 23 Mar 2026   AIC                           -278.587
Time:                        14:45:52   BIC                           -214.208
Sample:                    02-01-1948   HQIC                          -253.756
                         - 03-01-2009
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0075      0.013      0.571      0.568      -0.018       0.033
ar.L1          0.0104      0.026      0.401      0.688      -0.041       0.061
ar.L2          0.2238      0.037      6.053      0.000       0.151       0.296
ar.L3          0.1613      0.037      4.410      0.000       0.090       0.233
ar.L4          0.1018      0.043      2.374      0.018       0.018       0.186
ar.L5          0.1230      0.035      3.530      0.000       0.055       0.191
ar.L6         -0.0078      0.033     -0.236      0.813      -0.073       0.057
ar.L7         -0.0353      0.044     -0.804      0.422      -0.121       0.051
ar.L8          0.0123      0.040      0.309      0.758      -0.066       0.091
ar.L9          0.0018      0.038      0.046      0.963      -0.073       0.076
ar.L10        -0.1116      0.038     -2.937      0.003      -0.186      -0.037
ar.L11         0.0289      0.036      0.791      0.429      -0.043       0.100
ar.L12        -0.1220      0.039     -3.158      0.002      -0.198      -0.046
sigma2         0.0385      0.001     27.878      0.000       0.036       0.041
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):              1237.71
Prob(Q):                              0.98   Prob(JB):                         0.00
Heteroskedasticity (H):               0.31   Skew:                            -0.13
Prob(H) (two-sided):                  0.00   Kurtosis:                         9.36
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-ste
p).
```

In [27]:

```
sigma2_a = fit_full.params['sigma2']
print(f"Residual variance (sigma^2_a): {sigma2_a:.6f}")
print(f"Residual std (sigma_a): {np.sqrt(sigma2_a):.6f}")
```

```
Residual variance (sigma^2_a): 0.038515
Residual std (sigma_a): 0.196252
```

In [28]:

```
params = fit_full.params
stderr = fit_full.bse

print("Parameter estimates and standard errors:")
print("-" * 50)
for name in params.index:
    print(f"{name:12s}: estimate = {params[name]:10.6f},  std err = {stderr[name]:
```

```
Parameter estimates and standard errors:
--------------------------------------------------
const       : estimate =   0.007485,  std err =   0.013120
ar.L1       : estimate =   0.010422,  std err =   0.025993
ar.L2       : estimate =   0.223836,  std err =   0.036980
ar.L3       : estimate =   0.161333,  std err =   0.036587
ar.L4       : estimate =   0.101795,  std err =   0.042872
ar.L5       : estimate =   0.122998,  std err =   0.034847
ar.L6       : estimate =  -0.007795,  std err =   0.033032
ar.L7       : estimate =  -0.035289,  std err =   0.043908
ar.L8       : estimate =   0.012339,  std err =   0.039985
ar.L9       : estimate =   0.001763,  std err =   0.038065
ar.L10      : estimate =  -0.111561,  std err =   0.037980
ar.L11      : estimate =   0.028864,  std err =   0.036492
ar.L12      : estimate =  -0.122021,  std err =   0.038643
sigma2      : estimate =   0.038515,  std err =   0.001382
```

Part (b): Forecast April–July 2009

In [29]:
```python
# Forecast 4 months: Apr, May, Jun, Jul 2009
forecast_b = fit_full.get_forecast(steps=4)
pred_mean_b = forecast_b.predicted_mean
pred_ci_b = forecast_b.conf_int(alpha=0.05)

months_b = ['2009-04', '2009-05', '2009-06', '2009-07']
results_b = pd.DataFrame({
    'Month': months_b,
    'Forecast': pred_mean_b.values,
    'Lower 95%': pred_ci_b.iloc[:, 0].values,
    'Upper 95%': pred_ci_b.iloc[:, 1].values
})
results_b.index = pred_mean_b.index

print("Forecasts for April–July 2009 (yt = change in unemployment rate):")
print(results_b.to_string())
```

```
Forecasts for April–July 2009 (yt = change in unemployment rate):
              Month  Forecast  Lower 95%  Upper 95%
2009-04-01  2009-04  0.267050  -0.117597   0.651697
2009-05-01  2009-05  0.175531  -0.209137   0.560199
2009-06-01  2009-06  0.172142  -0.222053   0.566337
2009-07-01  2009-07  0.136297  -0.263036   0.535629
```

Part (c): Refit with data until December 2007

In [30]:
```python
yt_dec2007 = yt.loc[yt.index <= '2007-12']
model_dec2007 = ARIMA(yt_dec2007, order=(12, 0, 0))
fit_dec2007 = model_dec2007.fit()

forecast_15 = fit_dec2007.get_forecast(steps=15)
pred_mean_15 = forecast_15.predicted_mean
pred_ci_15 = forecast_15.conf_int(alpha=0.05)

print("Forecast origin: December 2007")
print("15-month forecasts (Jan 2008 – Mar 2009):")
results_15 = pd.DataFrame({
```

```
    'Forecast': pred_mean_15.values,
    'Lower 95%': pred_ci_15.iloc[:, 0].values,
    'Upper 95%': pred_ci_15.iloc[:, 1].values
}, index=pred_mean_15.index)
print(results_15)
```

```
Forecast origin: December 2007
15-month forecasts (Jan 2008 - Mar 2009):
            Forecast  Lower 95%  Upper 95%
2008-01-01 -0.014453  -0.398097   0.369191
2008-02-01  0.078349  -0.305296   0.461993
2008-03-01  0.069680  -0.322301   0.461661
2008-04-01  0.004499  -0.391817   0.400815
2008-05-01  0.080780  -0.319293   0.480853
2008-06-01 -0.007792  -0.413881   0.398297
2008-07-01  0.012367  -0.394548   0.419282
2008-08-01  0.027633  -0.379914   0.435180
2008-09-01 -0.004811  -0.413706   0.404084
2008-10-01 -0.016666  -0.425975   0.392643
2008-11-01  0.014117  -0.395817   0.424050
2008-12-01 -0.048254  -0.458746   0.362238
2009-01-01  0.002231  -0.411977   0.416438
2009-02-01 -0.018001  -0.432213   0.396212
2009-03-01 -0.024249  -0.439152   0.390654
```

Part (d): Plot true observations, forecasts, and prediction intervals

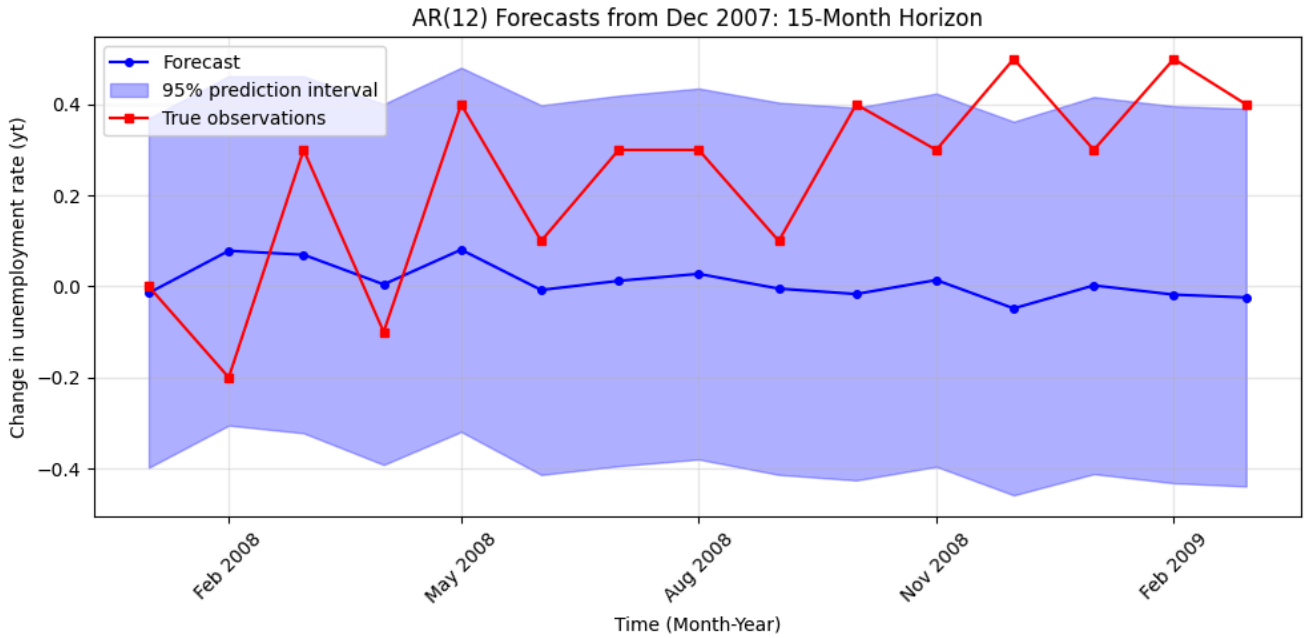
In [31]:
```
actual_15 = yt.loc['2008-01':'2009-03']
fig, ax = plt.subplots(figsize=(10, 5))

# Forecast
x_dates = pred_mean_15.index
ax.plot(x_dates, pred_mean_15.values, 'b-o', label='Forecast', markersize=4)
ax.fill_between(x_dates, pred_ci_15.iloc[:, 0], pred_ci_15.iloc[:, 1], alpha=0.3,

# Actual
ax.plot(actual_15.index, actual_15.values, 'r-s', label='True observations', marke

ax.set_xlabel('Time (Month-Year)')
ax.set_ylabel('Change in unemployment rate (yt)')
ax.set_title('AR(12) Forecasts from Dec 2007: 15-Month Horizon')
ax.legend(loc='upper left')
ax.grid(True, alpha=0.3)

# Date format
ax.xaxis.set_major_formatter(mdates.DateFormatter('%b %Y'))
ax.xaxis.set_major_locator(mdates.MonthLocator(interval=3))
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```


AR(12) Forecasts from Dec 2007: 15-Month Horizon
Forecast
95% prediction interval
True observations
Change in unemployment rate (yt)
0.4
0.2
0.0
−0.2
−0.4
Feb 2008
May 2008
Aug 2008
Nov 2008
Feb 2009
Time (Month-Year)